

Total no. of Pages:2

Roll. No:.....

IIIrd SEMESTER, B. TECH: ECE (VLSI Design and Technology)
END-SEMESTER THEORY EXAMINATION, NOV-DEC, 2025
Course Code: VTECC304 Course Title: Microelectronics
Time: 3:00 Hours Max. Marks: 50
Note: Missing data/information, if any, may be suitably assumed and mentioned in the answer.

Q.No.	Question	Marks	CO
1	Attempt any two parts		
(a)	Draw the high frequency model of a MOSFET in common source configuration and determine the expression for its short circuit current gain and the expression for the unity gain frequency therefrom, in terms of the relevant parameters.	5	CO1
(b)	For the differential amplifier, shown in Fig. 1 drive an expression for its small signal transconductance gain and I_{out} .	5	CO1
(c)	For the compensated common source amplifier of Fig. 2 carry out an analysis, using square-law, to determine the DC and AC components appearing in the output voltage of the circuit.	5	CO1

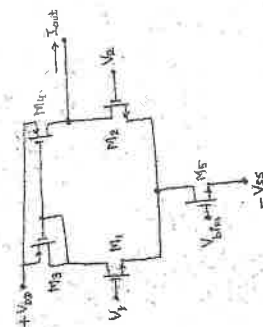


Fig.1

2.	Attempt any two parts			
(a)	For the CS amplifier of Fig. 3 draw a high frequency model and by appropriate analysis show the influence of Miller's effect in the expression for the 3-dB bandwidth of the circuit.	5	CO2	
(b)	Explain how in place of the resistor R_D in the circuit of Fig. 3, a MOS current mirror can be used as an active load.	5	CO2	
(c)	Draw the circuit of a cascode amplifier, biased by a constant current source and explain how cascoding results in an increased value of the voltage gain.	5	CO2	

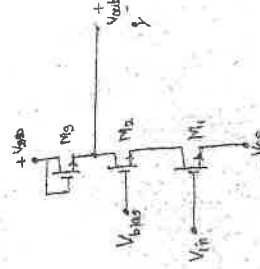


Fig. 2

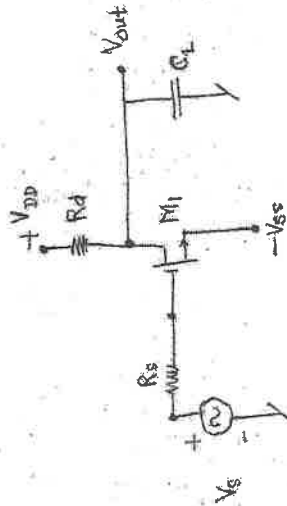
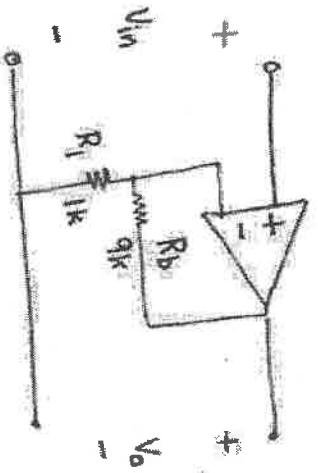


Fig.3

3.	Attempt any two parts		
(a)	For the non-inverting amplifier shown in Fig.4, the open loop gain 'A' of the op-amp is $A \cong \frac{\omega_t}{s}$ where ω_t is the gain bandwidth product equal to $2\pi \times 10^6$. Determine the bandwidth of the close-loop amplifier by deriving the formula used.	5	CO3
 Fig. 4			
(b)	Draw the block diagram of a series-series feedback amplifier and determine the expressions for its closed loop gain, input impedance and output impedance.	5	CO3
(c)	Analyze the RC-phase oscillator shown in Fig. 5 and determine the expressions for the condition of oscillation and frequency of oscillation, in terms of relevant parameters, If $R = 10\text{ K}\Omega$, $C = 0.01\mu\text{F}$, $R_D = 10\text{ K}\Omega$ and r_o of the MOSFET be $100\text{ K}\Omega$, what should be the value of g_m of the CS amplifier so that the circuit can generate sinusoidal oscillations and what would be the frequency of the generated sine wave?	5	CO3

4.	Attempt any two parts		
(a)	Compare the simple, Widlar, Wilson and Cascode current mirrors, in terms of the relevant features.	5	CO4
(b)	Distinguish between class-A, class-B and class-AB of operations as applicable to power amplifiers.	5	CO4
(c)	What is the cause of crossover distortion in a push-pull amplifier? Suggest and explain a method to reduce this distortion.	5	CO4
5.	Attempt any two parts		
(a)	What are the various steps involved in the preparation of a wafer in the IC fabrication process?	5	CO5
(b)	For what purpose diffusion and ion-implantation are used?	5	CO5
(c)	What is the function of Photolithography in IC fabrication process?	5	CO5

